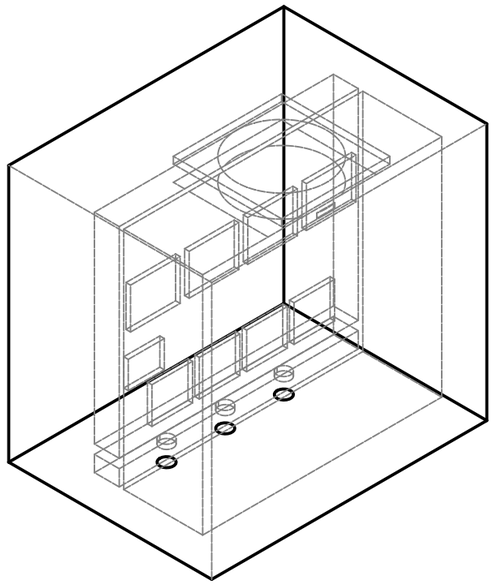
ASSEMBLY – OpenSmell v1.1 sensing platform

isometric general arrangement – shell, baffle plane, 3× nostril, filter pack, PCB floor, DIA40 rear exhaust

BILL OF MATERIALS

ITEM	PART / DESCRIPTION	QTY	MATERIAL / SOURCE	TYPE
1	Shell 122×114×90, cavity 106×100×48	1	PETG FDM (or AL 6061-T6)	make
2	Baffle plate 106×100×2, 8 wells + ref port	1	PTFE machined (or PVDF)	make
3	Axial fan DIA40 × t10, 5 V	1	off-the-shelf (substitute any 40mm)	buy
4	Filter pack PTFE, full-flow 106×11 face	1	pleated, ≥100 cm² media	buy
5	Main PCB 114×105 (SHT31, MCU)	1	FR4 1.6 mm + build	make

geometry from real HLR of the v1.1 solid; all dims mm; per blueprint.md 0.4

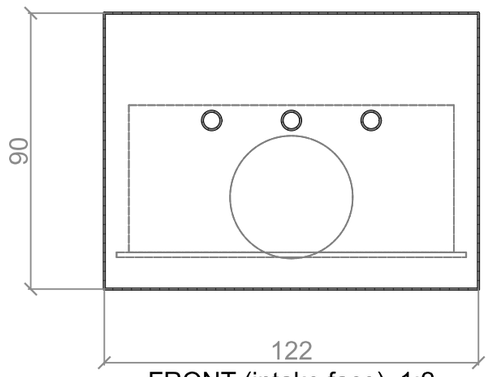


ISO – general arrangement (NTS)

GENERAL TOLERANCE ISO 2768-mK · PROJECT OpenSmell v1.1 REV A
NOTES: all dims mm; deburr/de-flash; seal wetted faces; print thin parts 0.2 mm layer

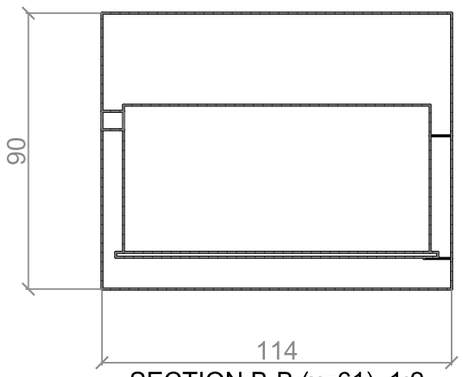
ASSEMBLY			SCALE NTS
DRAWING No. OS-000	REV A	SHEET 1/3	SIZE A3 UNITS mm
MATERIAL see BOM FINISH see BOM			FIRST ANGLE (ISO 128)
DRAWN OpenSmell	CHECK –	DATE 2026-10-16	

SHELL – main body (OS-001)

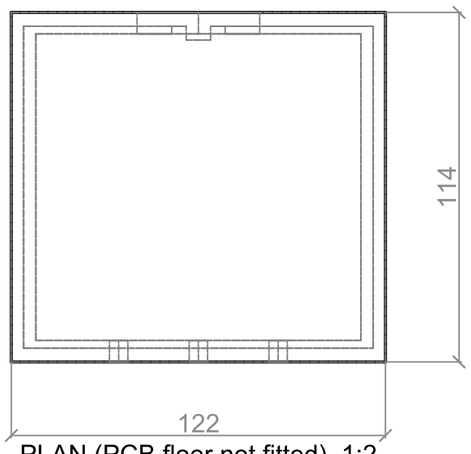


FRONT (intake face) 1:2

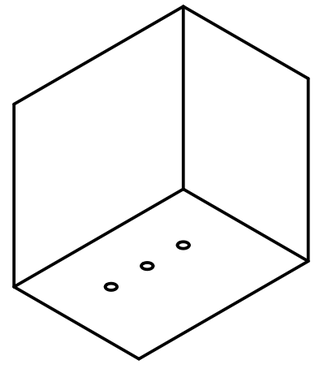
3×DIA6 nostrils



SECTION B-B (x=61) 1:2



PLAN (PCB floor not fitted) 1:2



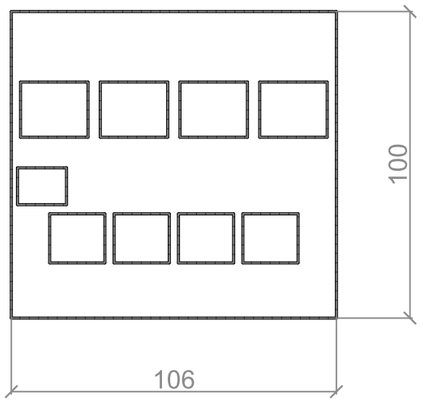
ISO – shell / cavity / baffle plane (NTS)

GENERAL TOLERANCE ISO 2768-mK · PROJECT OpenSmell v1.1 REV A
NOTES: all dims mm; deburr/de-flash; seal wetted faces; print thin parts 0.2 mm layer

SHELL – main body			SCALE 1:2
DRAWING No. OS-001	REV A	SHEET 2/3	SIZE A3 UNITS mm
MATERIAL PETG, FDM 0.2 mm, 4 perimeters, vapor-sealed inner cavity FINISH as-printed, de-flash; wetted faces seamed/sealed			FIRST ANGLE (ISO 128)
DRAWN OpenSmell	CHECK –	DATE 2026-10-16	

BAFFLE – well plate (OS-002)

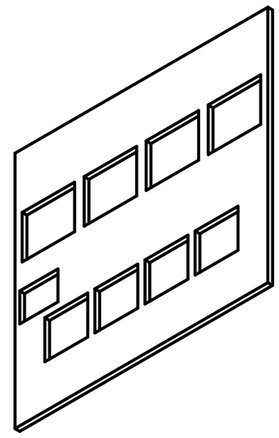
pitch 26 analog / 21 digital



PLAN (8 wells + ref port) 1:2



SECTION A-A (plate t=2, wells open-bottom) 1:1



ISO – wells open-bottom (NTS)

GENERAL TOLERANCE ISO 2768-mK · PROJECT OpenSmell v1.1 REV A
NOTES: all dims mm; deburr/de-flash; seal wetted faces; print thin parts 0.2 mm layer

BAFFLE – well plate			SCALE 1:2
DRAWING No. OS-002	REV A	SHEET 3/3	SIZE A3 UNITS mm
MATERIAL PTFE, machined (inert, non-porous, hydrophobic) FINISH deburr; 8 open-bottom wells + 1 sensor port, d/w to CONST			FIRST ANGLE (ISO 128)
DRAWN OpenSmell	CHECK –	DATE 2026-10-16	